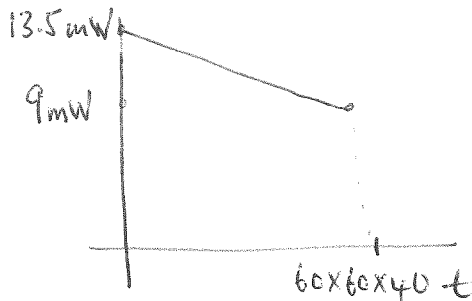


MEMORANDUM

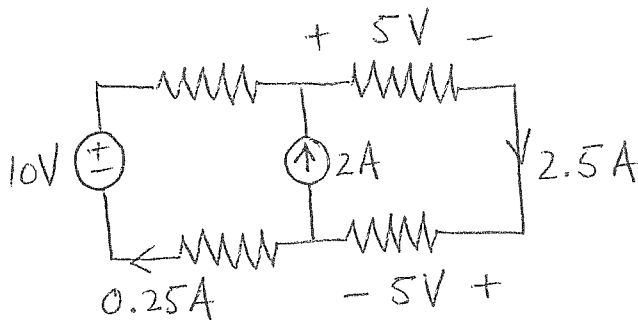
EBN122 Klastoets/Class Test#1: 7 Aug 2013

- Q1 Die vervaardiger van 'n 1.5V flitslig sê dat die battery vir 40 ure kontinu 9 mA stroom sal lewer. Gedurende daardie tyd val die spanning van 1.5V tot 1.0V (aanvaar dat die spanning lineêr met tyd val). Hoeveel energie lewer die battery in die 40 uur tyd-interval?
The manufacturer of a 1.5V flashlight says that the battery will deliver 9 mA current for 40 continuous hours. During that time the voltage will drop from 1.5V to 1.0V (assume the voltage drop is linear with time). How much energy does the battery deliver in the 40 hour time interval? [2]



$$W = 9 \times 10^{-3} \times 60 \times 60 \times 40 + \frac{1}{2} \times 4.5 \times 10^{-3} \times 60 \times 60 \times 40 = 1296 + 324 = 1620 \text{ J} \checkmark$$

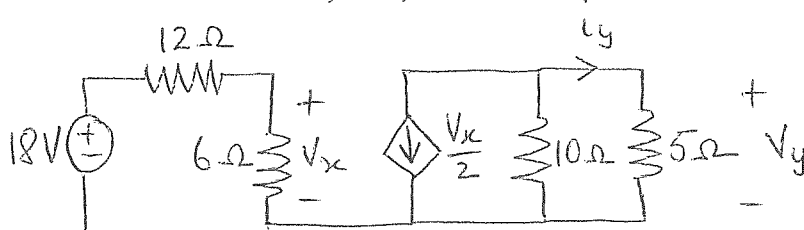
- Q2 Bereken die drywing ge-assosieer met die 10V spanningsbron en die 2A stroombron. Calculate the power associated with the 10V voltage source and the 2A current source. [2]



$$P_{10V} = -0.25 \times 10 = -2.5 \text{ W} \checkmark$$

$$P_{2A} = -2 \times 10 = -20 \text{ W} \checkmark$$

- Q3 Bereken v_x , v_y en i_y in die stroombaan. Calculate v_x , v_y and i_y in the circuit. [3]

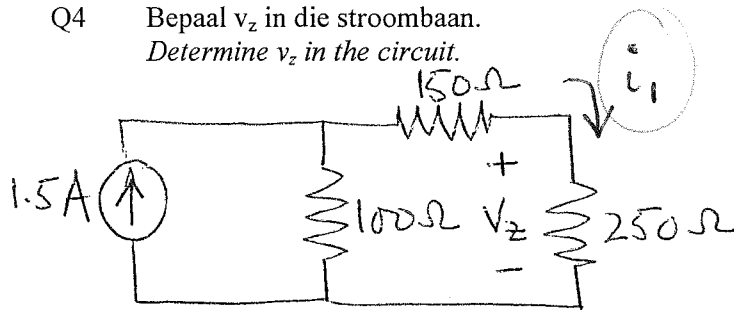


$$V_x = \frac{6}{6+12} \times 18 = 6 \text{ V} \checkmark$$

$$i_y = -\frac{10}{10+5} \times 3 = -2 \text{ A} \checkmark$$

$$V_y = -2 \times 5 = -10 \text{ V} \checkmark$$

Q4 Bepaal v_z in die stroombaan.
Determine v_z in the circuit.

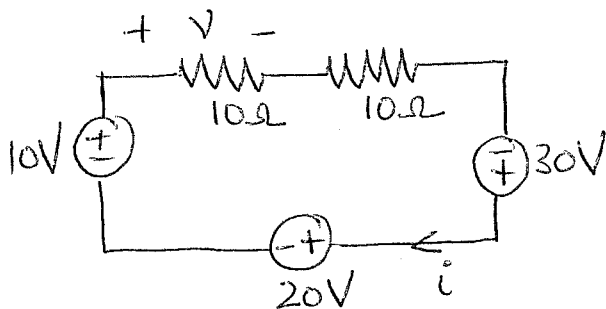


$$i_1 = \frac{100}{100 + 400} \times 1.5$$

$$= 0.3 \text{ A}$$

$$V_z = 0.3 \times 250 = 75 \text{ V}$$

Q5 Bereken v en i .
Calculate v and i .

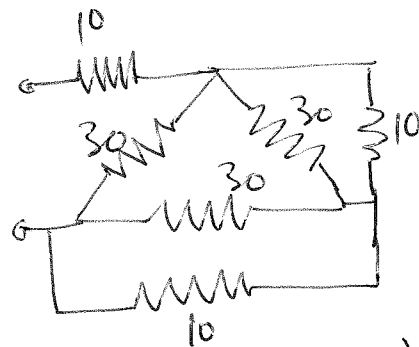
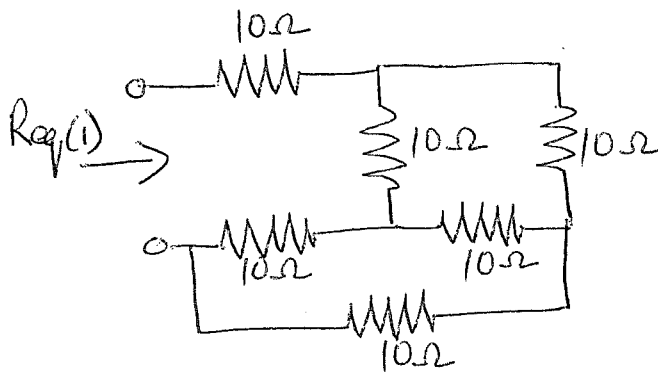


$$-10 + v + v - 30 + 20 = 0$$

$$v = 10 \text{ V}$$

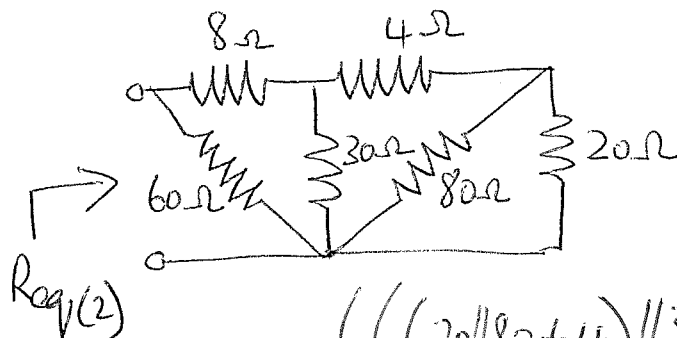
$$i = \frac{10 - 20 + 30}{20} = 1 \text{ A}$$

Q6 Bereken R_{eq} vir elkeen van die bane.
Calculate R_{eq} for each of the circuits.



$$R_{eq(1)} = ((30 \parallel 10) + (30 \parallel 10)) \parallel 30 + 10$$

$$= 20 \Omega$$



$$R_{eq(2)} = 15 \Omega$$

$$(((20 \parallel 80 + 4) \parallel 30) + 8) \parallel 60$$